

Sequential Behavior Modeling and Simulation for Marketplace AI Evaluation

Research proposal for DoorDash AI Research Fellowship | Yulong Zhao, Statistics PhD, University of Arizona

Motivation and Question

Many AI problems in a local-commerce marketplace are sequential rather than static. A consumer searches, compares, revises a choice, abandons, or completes an order; merchants and couriers respond under timing, capacity, and information constraints. My proposed direction is to build a statistical machine learning framework for modeling and simulating these behavioral event sequences. The central question is: when a recommendation, search result, incentive, or assistant changes the information shown to a user or merchant, what behavioral shifts should we expect, and where is the model uncertain?

Research Direction

I would study marketplace interactions as variable-length event sequences with latent behavioral states. Depending on the problem, these states may capture intent, urgency, price sensitivity, friction, or uncertainty. The framework would have three parts:

- **Sequence representation:** preserve order, timing, context, and outcomes rather than collapsing behavior into aggregate counts.
- **Latent behavior modeling:** estimate interpretable states and time-varying action effects with sparse or regularized structure.
- **Simulation-based evaluation:** generate semi-synthetic sequences to stress-test AI policies under sparsity, delayed outcomes, rare user states, and distribution shift.

The methodological principle is that prediction alone is not enough. A useful model should expose failure modes, quantify uncertainty, and support careful comparison between candidate systems. In a marketplace setting, this means asking whether the implied behavior is plausible across users, merchants, time windows, and operational constraints.

Internship Plan

In a 12-week internship, I would keep the scope focused. First, I would define one concrete marketplace sequence problem with mentors, such as user search-to-order behavior, merchant response patterns, or evaluation of an AI-assisted recommendation surface. Second, I would compare baseline sequence features with a latent-state or regularized sequence model. Third, I would build diagnostics: holdout evaluation, subgroup checks, uncertainty summaries, and semi-synthetic stress tests. The deliverable would be a reproducible prototype and a concise research memo describing where the method works, where it fails, and what would be needed for production-grade evaluation.

Fit with My Background and DoorDash

My current PhD work is closely related to this direction. In scalar-on-process research with Prof. Xueying Tang, I model variable-length action logs, separating action frequency from timing effects and estimating interpretable time-varying behavior patterns. In my additive cognitive diagnosis modeling project, I am building a Python research codebase that separates simulation, estimation, support selection, refitting, and evaluation so that methods can be compared cleanly.

DoorDash is a strong setting for this work because its AI systems sit inside a real marketplace where user behavior, merchant behavior, timing, and constraints interact. I would bring a statistics-first perspective: build models that are predictive, diagnosable, simulation-aware, and honest about uncertainty.